Ex.No:12 End-to-End communication at Transport layer

8.10.26

a) Implement echo client server using TCP/ UDP sockets.

TCP

```
import socket
import threading
def startServer():
    server = socket.socket(AF_INET)
    server.bind('127.0.0.1',5000)
    server.listen()
    print("[SERVER] listening")
    conn,addr = server.accept()
    print(f"[SERVER] connected")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() =='quit':
            print("[SERVER] connection closed by client")
            break
        print(Received)
    conn.close()
    Server.close()
def client():
    client = socket.socket(AF_INET)
    client.connect(('127.0.0.1',5000))
    print("Connect to Server")
```

```
while True:
    msg = input("Enter")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print("Server Response")
if __name__ == "__main__":
    server_thread = threading.Thread(
        target = server, daemon = True)
    server_thread.start()
    client().
```

Output:

```
[SERVER] Listening on 127.0.0.1:5000
Connected to server.
Enter message : hello
Received : hello
```

b) UDP:

```
import socket
import threading
def udp_server():
    s = socket.socket(AF_INET)
    s.bind('127.0.0.1', 12000)
    print("Running")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
```

```
            print("client ended the connection")
            break
        print("Server Received")
        s.sendto(data, addr)
    s.close()
    print("Server stopped.")

def udp_client():
    c = socket.socket(AF_INET)
    Server = ('127.0.0.1', 12000)
    while True:
        msg = input()
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("closing connection")
            break
        data, - = c.recvfrom(1024)
        print("data.decode()")
    c.close()
    print(client stopped)

if __name__ == "__main__":
    threading.Thread(target = udp-
                     server,
        daemon = True).start() udp_client()
```

Output:

Server received from ('127.0.0.1', 49670)

hi

Enter msg: Hello

Server received : hello

Echo from server : hello

Enter msg: How are you

Server Received : how are you

Result:

The echo server programs were successfully implemented using TCP and UDP sockets, and message transmission and reception were verified.